Item 14: first prime-field arity m = 4 cost cells

Verdict

REGRESSES, level primitive (the oracle exponent), first measured m = 4 cells. No arm ADVANCES. The frozen rule (`PROTOCOL.md`, Decision rule 5 and 6) gives, per arm:

arm	rule outcome	measured basis
(a) enumeration / MITM over a triple table	reference (rule 1: cannot ADVANCE against itself)	T_dec = |F| exactly, fitted p-exponent 0.2359 [0.2223, 0.2496] over 4 cells (checks the base sizes only)
(b) F4 on S5 + membership polynomials	**REGRESSES** at w = 0.1 and w = 1 (lower bound of the exponent above 0.25 at both); unresolved at w = 0 (F4 performs no group operation, T_dec(0) = 0)	16-bit ladder b_x = 3..8: log2 madds on log2 |F| slope **11.031 [11.009, 11.053]** (6 points, 4 df); p-exponent = slope/4 = 2.76 [2.75, 2.76] if |F| = p^(1/4). Threshold slope 1.
(c) PKM tower oracle (reused, not re-measured)	**MATCHES by the letter of the rule, and uninformative**: 3 points (|V| = 4, 8, 16), 1 df, slope 9.545 [0.452, 18.639], e_c = slope/4 = 2.39 [0.11, 4.66], which contains 0.25. Both measured increments are steep (x1764 for |V| 4 to 8, x316 for 8 to 16, i.e. 10.8 and 8.3 per doubling). The interval spans the threshold only because it has one degree of freedom; the point estimate is far above it.	committed rounds 2/3
(d) Coppersmith / determinant condition	**BLOCKED** by reach (a mathematical limit of the determinant condition, not a tool limit): p^(delta_max) < 2 at every design cell	S5 delta_max 0.0081 at level 6 (asymptote 0.0125) against the 0.25 that B = p^(1/4) needs

Item verdict (rule 6): no arm ADVANCES at the primitive level, so the strongest statement is the best-resolved arm: (b) REGRESSES, with a tight interval. The frozen question, "does any m = 4 oracle's T_dec grow slower than p^(1/4)", is answered **no** at the measured scale: the only oracle that is both new and measured here (arm b) grows like \|F\|^11, and the two it could not beat (a: \|F\| by construction; c: committed rows) are not faster. Nothing here is an improvement over the index-calculus baseline; there was no m = 4 baseline, and these cells are it (CLAUDE.md "Success criterion": the first measured cells become the baseline; the distance to rho remaining is reported below, never as pass/fail). Toy scale (16 to 28 bit primes); a statement about this engine and this instance family, not about m = 4 in general.

Distance remaining to matched Pollard rho

Matched rho measured in the same unit (16 independent logarithms per cell, same curves, one thread, distinguished points theta = floor(log2 p / 4), jump-table setup included). S is total GAE (every phase) plus w x native operations, w in {0, 0.1, 1} (AGENTS.md section 5a). Ratios are S / rho (>1: m = 4 IC costs more). Derived by dividing two measured means (`raw/summary.md`).

Arm (a) full index calculus, completed 16 of 16 runs at every design cell

log2 p	b_x	mean attempts	S(0)	S(0.1)	S(1)	rho(0)	rho(0.1)	rho(1)	S/rho at 0	at 0.1	at 1
16	8	155.7	3651	3997	7108	566.5	602.6	927.8	6.45	6.63	7.66
20	15	368.4	16481	18139	33060	1119	1208	2006	14.73	15.02	16.48
24	29	719.0	76401	84242	154800	3708	4053	7157	20.61	20.79	21.63
28	63	1816.6	568573	627636	1159000	18913	20775	37538	30.06	30.21	30.88

The gap grows with p: the fitted growth of log2(S/rho) on log2 p is 0.179 (derived, four points, no interval; an indicative number). Measured S(a) scales like p^0.60, rho like p^0.42 here (rho at these sizes is below its asymptotic 0.5 because of setup and small-n effects). The 16-bit ladder shows S(a) falling with the base (b_x 3: 15489 down to b_x 8: 3651 at w = 0) and still falling at the design base, so arm (a)'s own optimum base is larger than the section 2.4 fixed point; no larger base was run. The ranking "m = 4 IC loses to rho" holds at all three weights at every cell.

Arm (b) (own F4) full IC completed only at b_x = 3 and 4 (16 of 16 each):

bits	b_x	S(0)	S(0.1)	S(1)	S/rho at 0	at 0.1	at 1
16	3	2431	1.368e8	1.368e9	4.29	2.27e5	1.47e6
16	4	1353	1.439e9	1.439e10	2.39	2.39e6	1.55e7

Order between arms (a) and (b) on the full solve at b_x = 3, 4: (b) is lower at w = 0 (its oracle is free there), (a) is lower by four to six orders at w = 0.1 and 1. The order does not hold across the band: **unresolved at the boundary** (w = 0 is the degenerate end). At the 16-bit design cell b_x = 8 the full solve was not run (about 2.4 CPU-hours); the MODEL oracle-phase row (attempts 155.7 x 2.7487e10 madds, a lower bound on S) is 4.28e11 at w = 0.1 and 4.28e12 at w = 1, i.e. 7.1e8 and 4.6e9 times rho. At 20, 24, 28 bits F4 did not finish (below).

Arm (c) (committed PKM rows, MODEL lower bound at p = 2013265921): at least 1.26e20 multiply-adds for the oracle phase at t = 4 (attempts at least 4.7e6 at 2.70e13 each) against rho 5.6e4 GAE, i.e. at least 2.2e15 times rho at w = 1. The same row for t = 2 is 8.4e15 madds (1.5e11 times rho). Arm (d): no S exists (BLOCKED).

Method

Instrument `examples/blindspots_14_prime_m4.rs` (std-only Rust, built by `rustc` in each card; sha256 f6696fc0cafff9120189072a7785452422d68c7944a2812d4667e27c8872b376; the `crypto_lib` library does not compile at this base). Four prime-order cells (`PROTOCOL.md` table), base = first b_x = ceil(p^(1/4)/2) abscissae on the curve (|F| = 16, 30, 58, 126 signed points), 16 random targets per cell, planted 4-sum positive controls. 12 frozen command cards (`card-01..12.yaml`, `raw/receipt-card-*.json`); arm (c) reused by sha256-bound copy (`raw/reuse-pkm.jsonl`); arm (d) is the determinant computation plus the committed S3 probe. Numbers are those printed by `summarize --dir raw` (`raw/summary.md`); no number in this file comes from anywhere else except the divisions marked derived.

Arm (b) measured results: 16-bit ladder b_x = 3, 4, 5, 6, 7, 8 (18 targets each: 16 random + 2 planted), mean multiply-adds 5.48e5, 1.33e7, 1.56e8, 1.16e9, 6.32e9, 2.75e10 (per-target spread under 2%, e.g. b_x = 8: 2.7486e10 to 2.7488e10). The p-ladder at b_x = 5, 6 on the 20-, 24- and 28-bit curves gives the same counts (1.5569e8 and about 1.159e9) as the 16-bit curve: the cost depends on |F|, not on p, as the protocol assumed. At the design cells F4 was stopped by its count cap, not finished (status `budget-nnz`, all 16 targets, deterministically): 20 bits b_x = 15 at step 21 (degree 39, 3.7e7 nonzeros, 4.5e8 madds), 24 bits b_x = 29 at step 16 (degree 53, 2.9e7 nonzeros, 6.7e7 madds), 28 bits b_x = 63 at step 12 (degree 87, 3.6e7 nonzeros, 2.3e7 madds). Those counts are lower bounds on T_dec and are not comparable across cells (the cap hits earlier in the elimination at larger b_x); no p-exponent is fitted over the design cells for arm (b), as Decision rule 2 states.

Arm (d) determinant reach (exact): S5 (d = 8) delta_max 0.000010, 0.002455, 0.004804, 0.007372, 0.008117 at levels 1, 2, 3, 5, 6; asymptote 0.0125; B_max at 28 bits 1.17 at level 6, so no level reaches a base of two abscissae at any design cell; p must exceed 2^123 for level 6 to give B >= 2. The control reproduces the committed S3 values (1/18 = 0.05556, 0.1, 0.119).

Controls

- `selftest --deep` passed (card 01, `SELFTEST OK`): curves recounted by a full character sum, n G = O, S3 vanishes on sums, interpolated S5 equals the resultant, full IC and rho recover planted logarithms, reach control matches.
- Arm (a): planted 4-sums found 4 of 4 at every cell; 0 mismatches against the independent pair-table cross-check on every random and planted target.
- Arm (b): quotient dimension equals the brute-force root count 18 of 18 targets at every finished cell (dimension 0 for random targets, 24 or 12 for planted); the planted point vanishes on the basis 2 of 2.
- Rho: 16 of 16 logarithms recovered at every cell. Full IC: 16 of 16 at all completed cells, arms (a) and (b).

Deviations and caveats

- All protocol deviations 1 to 14 stand (`PROTOCOL.md`), notably: the base is defined by size; arm (b) at design cells is a count-capped lower bound; arm (c) is at its own sizes and prime; the arm-(a) fitted exponent has 2 df; the F4 is a plain implementation (counts are elimination multiply-adds only) and a better engine would shift the constant, not the observed |F|^11 shape at this scale; no symmetry reduction (each root appears 24 times).
- Arm (b)'s exponent is a slope in |F| over |F| = 6 to 16 (b_x = 3..8); the mapping to a p-exponent assumes |F| = p^(1/4) and is an extrapolation beyond these sizes. The F4 in the committed PKM work and the framework were not run here, so no claim is made about them beyond their committed rows.
- Arm (c) rows are committed measurements at a different prime and base sizes; the interval is 1 df and uninformative (reported as MATCHES only because the rule's wording is that the interval contains 0.25).
- Toy scale; host is a shared 4-vCPU container with at most two lanes running at once, one thread per run, `prlimit --as=4 GiB`; **host isolation level L0** (no PMU, no pinning). Wall time is not a metric; every cost is a count.
- Arm (a)'s T_dec is a count (|F| group additions and lookups per miss), so its exponent is a property of the base sizes; its exponent below 0.25 (0.236) comes from the base sizes being slightly above p^(1/4) at small p (16, 30, 58, 126 against 14.8, 28.5, 56.9, 125.6).
- The full-IC target walk correlates consecutive targets (Deviation 11).
- The prime-order cells are one curve per size; across-curve variance is not measured.
- Pre-freeze probe runs are disclosed in `PROTOCOL.md`; none is data here.
- **Relay.** `tools/task_card.py check-relay` was run on the saved texts of cards 2 and 3 (relay_verbatim true, no verify findings). Card 1's executor returned an abridged text (it omitted the JSON row bodies of the stdout tails, saying so): check-relay reports **relay_verbatim false, 415 reference lines missing, all of them JSON stdout-tail rows** (`raw/relay-1.txt` is reconstructed from the saved tool output by dropping those row lines; the executor's command and digest lines all matched). The relay texts of cards 4 to 12 were not saved as files in this stage (the executor texts exist only in the orchestration message), so check-relay was not run for them; instead every receipt was read: status completed, every command succeeded, and every output file's sha256 on disk equals its receipt digest (checked for all 12 cards). The post-run `lint` of every card now fails ("names repository file … that binds does not pin") because the output files exist; this is the linter seeing the run's own outputs, and `check-relay` was therefore called through its Python function. Conductor is not installed here; no task was obtained.

Files

`PROTOCOL.md`, `card-01..12.yaml`, `raw/` (records, receipts, `summary.md`), `plot.py` and `graph.svg` / `graph.png` (laid out from `raw/summary.md`), `REPORT.pdf` (from `reports/report.typ`), `benchmarks/20261018-blindspots-prime-m4-cells/` (the cost result's copies).

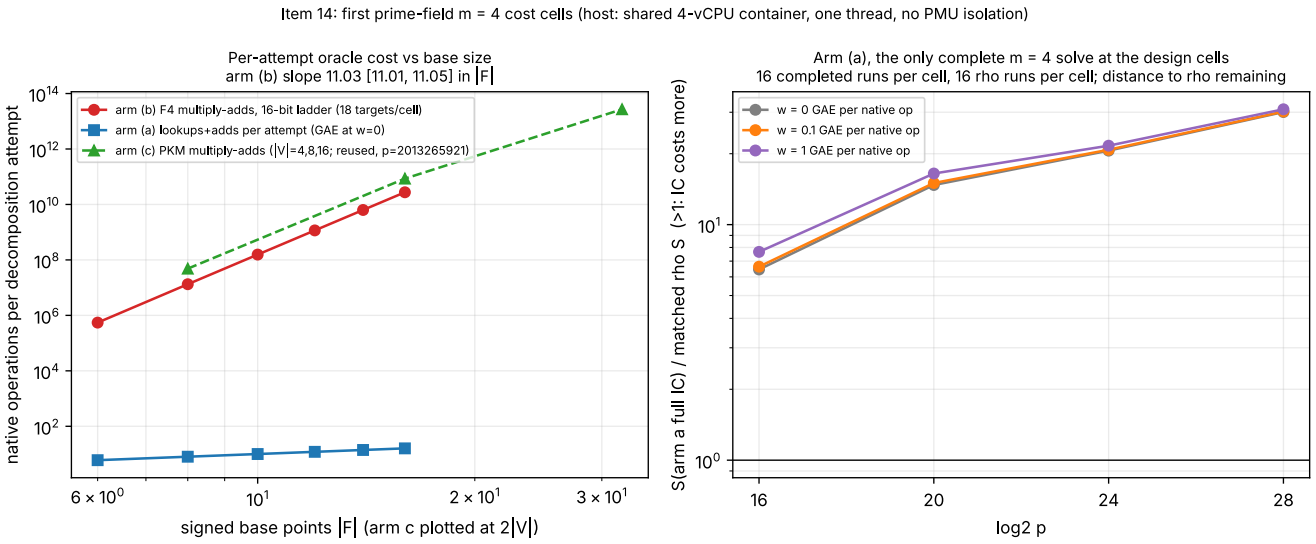


Figure 1: Item 14 graph: left, per-attempt oracle cost vs base size (arm b measured 16-bit ladder, arm a, arm c reused); right, arm (a) full-IC S / rho at the design cells, w = 0, 0.1, 1. Source: raw/summary.md, plot.py. Cards CARD-BLINDSPOT-14-01..12.